Brian D. Segal

Washington, D.C.

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EDUCATION

PhD in Biostatistics, University of Michigan, Ann Arbor, MI — Jul 2017

MS in Biostatistics, University of Michigan, Ann Arbor, MI — May 2013

BS in Biology, Virginia Tech, Blacksburg, VA, *summa cum laude* — May 2007

POSITIONS

Terion (formerly Indigo Ag), Boston, MA (Remote)
Staff Data Scientist — Jun 2026 – Present

Indigo Ag, Boston, MA (Remote)
Staff Data Scientist — Sep 2022 – May 2026
Senior Statistician — Jan 2021 – Aug 2022

Flatiron Health, New York, NY
Senior Quantitative Scientist — Mar 2020 – Dec 2020
Quantitative Scientist — Aug 2017 – Feb 2020

Google, Advanced Measurement Technologies, New York, NY
Intern — May 2016 – Aug 2016

University of Michigan, Department of Orthopaedic Surgery, Ann Arbor, MI
Research Associate II — Sep 2015 – Nov 2016

University of Michigan, Institute for Social Research, Ann Arbor, MI
Graduate Student Research Assistant — Sep 2014 – Aug 2015
Regents' Fellow — Sep 2013 – Aug 2014

University of Michigan, Department of Biostatistics, Ann Arbor, MI
Graduate Student Research Assistant — Jan 2012 – Aug 2013

Abt Associates, Environment and Resources Division, Bethesda, MD
Analyst — Jun 2010 – Mar 2011
Associate Analyst — Jun 2008 – May 2010
Research Assistant — Jul 2007 – May 2008

Virginia Tech, Department of Civil and Environmental Engineering, Blacksburg, VA
Research Assistant — Aug 2006 – May 2007

Environmental Protection Agency, Office of Pesticide Programs, Communication Services Branch, Crystal City, VA
Intern — May 2006 – Aug 2006

Virginia Tech, Department of Biological Sciences, Blacksburg, VA
Research Assistant — May 2005 – Aug 2005

PUBLICATIONS

Mathers, C., Black, C. K., **Segal, B. D.**, Gurung, R. B., Zhang, Y., Easter, M. J., Williams, S., Motew, M., Campbell, E. E., Brummitt, C. D., Paustian, K., Kumar, A. A. (2023). Validating DayCent-CR for cropland soil carbon offset reporting at a national scale. *Geoderma*, 438, 116647. https://doi.org/10.1016/j.geoderma.2023.116647.

Harton, J., **Segal, B. D.**, Mamtani, R., Mitra, N., Hubbard, R. A. (2023). Combining real-world and randomized control trial data using data-adaptive weighting via the on-trial score. *Statistics in Biopharmaceutical Research*, 15(2), 408–420. https://doi.org/10.1080/19466315.2022.2071982.

Tan, W. K., **Segal, B. D.**, Curtis, M. D., Baxi, S. S., Capra, W. B., Garrett-Mayer, E., Hobbs, B. P., Hong, D. S., Hubbard, R. A., Zhu, J., Sarkar, S., Samant, M. (2022). Augmenting control arms with real-world data for cancer trials: Hybrid control arm methods and considerations. *Contemporary Clinical Trials Communications*, 30, 101000. https://doi.org/10.1016/j.conctc.2022.101000.

Sondhi, A., **Segal, B. D.**, Snider, J., Humblet, O., McCusker, M. (2021). Bayesian additional evidence for decision making under small sample uncertainty. *BMC Medical Research Methodology*, 21(1), 221. https://doi.org/10.1186/s12874-021-01432-5.

Tan, W. K., Bryan, J., **Segal, B. D.**, Bellomo, L., Nussbaum, N., Tucker, M., Torres, A. Z., Bennette, C. S., Capra, W. B., Curtis, M. D., Miksad, R. A. (2021). Emulating Control Arms for Cancer Clinical Trials Using External Cohorts Created From Electronic Health Record-Derived Real-World Data. *Clinical Pharmacology & Therapeutics*, 111(1), 168–178. https://doi.org/10.1002/cpt.2351.

Segal, B. D. (2019). Toward replicability with confidence intervals for the exceedance probability. *The American Statistician*, 75(2), 128–138. https://doi.org/10.1080/00031305.2019.1678521.

Segal, B. D., Braun, T., Gonzalez, R., Elliott, M. R. (2019). Tests of matrix structure for construct validation. *Psychometrika*, 84(1), 65–83. https://doi.org/10.1007/s11336-018-9647-4 [open access version].

Segal, B. D., Elliott, M. R., Braun, T., Jiang, H. (2018). P-splines with an ℓ_1 penalty for repeated measures. *Electronic Journal of Statistics*, 12(2), 3554–3600. https://doi.org/10.1214/18-EJS1487.

Burgard, S. A., Lin, K. Y., **Segal, B. D.**, Elliott, M. R., Seelye, S. S. (2018). Stability and change in health behavior profiles of U.S. adults. *The Journals of Gerontology: Series B*, 75(3), 674–683. https://doi.org/10.1093/geronb/gby088.

Segal, B. D., Bennette, C. S. (2018). Re: "Transportability of trial results using inverse odds of sampling weights." *American Journal of Epidemiology*, 187(12), 2716–2717. https://doi.org/10.1093/aje/kwy190.

Segal, B. D., Braun, T., Elliott, M. R., Jiang, H. (2018). Fast approximation of small p-values in permutation tests by partitioning the permutations. *Biometrics*, 74(1), 196–206. https://doi.org/10.1111/biom.12731.

PATENTS

Campbell, E. E., et al. (24 inventors, including **Segal, B. D.**). Systems and methods for ecosystem credit recommendations. US Patent 11,830,089. Nov 28, 2023.

Segal, B. D., Brummitt, C. D. Addressing incomplete soil sample data in soil enrichment protocol projects. US Patent 11,592,431. Feb 28, 2023.

TALKS, POSTERS, AND ROUNDTABLES

Panel: "How decarbonization hits your dinner plate: Growing more rice with less methane." DC Climate Week, Apr 2026. Covered in FoodNavigator.

Talk: "Carbon quantification deep dive." Indigo Ag webinar, Mar 2023.

Talk: "Carbon quantification deep dive." Indigo Ag science and policy forum, Oct 2022.

Talk: "Hybrid control arms with RWD for cancer trials: Why, what, when, and how." JSM, Aug 2020.

Roundtable: "Case studies in the use of real world evidence to improve regulatory decision making." ASA biopharmaceutical section regulatory-industry workshop, Sep 2019.

Poster: "Tests of matrix structure for construct validation." American Psychological Association conference, Aug 2019.

Talk: "Complex data in, nuanced answers out: Lessons learned analyzing electronic health record data in oncology." JSM, Jul 2019.

Speed session (talk and poster): "Quantifying the number of events borrowed from external data in hybrid control arms." JSM, Jul 2019.

Talk: "Biostatistics and Flatiron Health: Harnessing the power of real world data through quantitative methods for cancer treatment, access, and care." University of Michigan symposium on big data, human health and statistics, Jul 2019.

Talk: "Exceedance probability for parameter estimates." JSM, Jul/Aug 2018.

Talk: "P-splines with an ℓ_1 penalty for repeated measures." Statistical learning and data science/nonparametric statistics conference, Jun 2018.

Poster (winner of best poster from Biostatistics department): "Tests of matrix structure for construct validation." Michigan Student Symposium for Interdisciplinary Statistical Sciences, Mar 2017.

Poster: "P-splines with an ℓ_1 penalty for repeated measures." ENAR, Mar 2017.

Talk (winner of travel award): "Fast approximation of small p-values in permutation tests by partitioning the permutation space." JSM Biometrics section student paper awards session, Aug 2016.

Poster: "Fast approximation of small p-values in permutation tests by partitioning the permutation space." ENAR, Mar 2016.

TECHNICAL REPORTS

Segal, B. D., Tan, W. K. (2020). A note on the amount of information borrowed from external data in hybrid controlled trials with time-to-event outcomes. https://arxiv.org/abs/2010.00433.

Amarakoon, S., Smith, J., **Segal, B. D.** (2013). Application of life-cycle assessment to nanoscale technology: Lithium-ion batteries for electric vehicles. US Environmental Protection Agency, Office of Pollution Prevention and Toxics, Design for the Environment Program. EPA 744-R-12-001.

Greco, S., Smith, J., **Segal, B. D.**, Post, E., Lynch, M., Hattis, D. (2009). Evaluating methods for quantifying human noncancer health risks: Case study application, draft report. US EPA.

Greco, S., Acquaye, A., Peak, K., **Segal, B. D.**, Rast, M. (2008). Framework for estimating costs and benefits associated with changes in the reference dose at federal facility hazardous waste sites. US EPA.

Segal, B. D. (2007). Biofilm forming properties of the ammonia oxidizing bacteria *Nitrosomonas europaea*. Undergraduate research project. Department of Civil and Environmental Engineering, Virginia Tech.

R PACKAGES

`exceedProb`: R package for computing confidence intervals for the exceedance probability of normally distributed estimators. Available at https://github.com/bdsegal/exceedProb.

`matrixStrucTest`: R package for testing symmetric matrices for block-diagonal structure under the null of exchangeable off-diagonal elements. Based on a permutation test with Hubert's gamma and a t-statistic. Available at https://github.com/bdsegal/matrixStrucTest.

`fastPerm`: R package for quickly approximating small permutation p-values for the difference and ratio of means. Available at https://github.com/bdsegal/fastPerm.

`gammaDist`: R package for computing the distribution and density of the difference of two gamma random variables under the null of equal distributions. Includes a saddlepoint approximation to the density. Available at https://github.com/bdsegal/gammaDist.

Prototypes

`psplinesl1`: R package for fitting additive mixed models with P-splines and an ℓ_1 penalty using alternating direction method of multipliers and cross validation. Available at https://github.com/bdsegal/psplinesl1.

TEACHING EXPERIENCE

Instructor: Statistical Programming Workshop, University of Michigan, Winter 2016.

Grader: Introduction to Public Health (PUBHLTH 610), University of Michigan, Fall 2014

Graduate Student Instructor: Introduction to Biostatistics (BIOSTAT 503), University of Michigan, Fall 2011

HONORS AND AWARDS

Best Poster from Biostatistics Department at Michigan Student Symposium for Interdisciplinary Statistical Sciences, "Tests of matrix structure for construct validation" 2017

Rackham Predoctoral Fellowship 2016

Travel Award for JSM, Biometrics Section, "Fast approximation of small p-values in permutation tests by partitioning the permutation space" 2016

Rackham Graduate Student Research Grant 2014

Regents' Fellowship 2013

Phi Beta Kappa 2005

Virginia Tech Paul Dirksen Smith Cycling Scholarship 2004

SERVICE AND MEMBERSHIPS

Reviewer for: *Annals of Applied Statistics, Biometrics, Clinical Trials, Journal of the American Statistical Association (Applications and Case Studies), Statistical Methods in Medical Research*

Session Chair, "Data science," JSM Jul/Aug 2018

Session Chair, "Statistical challenges in the analysis of EHR data," JSM Jul/Aug 2018

Member, Eastern North American Region International Biometric Society 2015 – 2021

Member, American Statistical Association 2012 – 2023

TECHNICAL SKILLS

Languages and tools: Python, R, Stan, SQL, AWS, Git, GitHub

AI coding agents: Claude Code, Codex, opencode